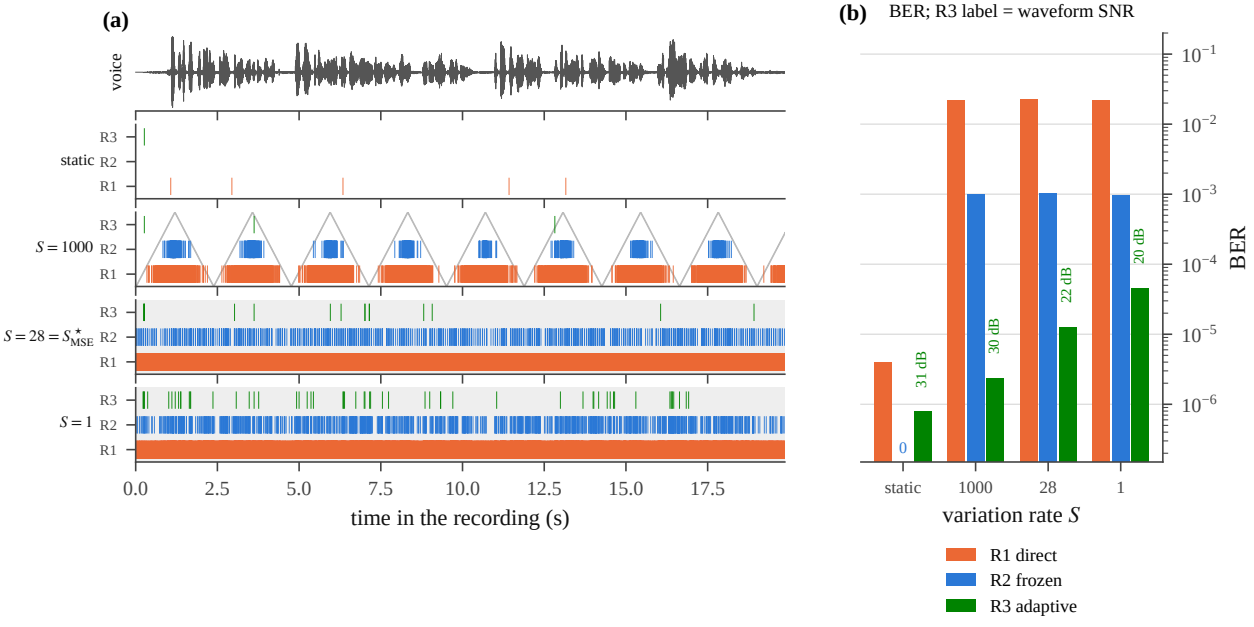

(a)
voice
static
R3
R2
R1
$S = 1000$
$S = 28 = S^{\star}_{\mathrm{MSE}}$
$S = 1$
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
time in the recording (s)
(b)
BER; R3 label = waveform SNR
$10^{-1}$
$10^{-2}$
$10^{-3}$
$10^{-4}$
$10^{-5}$
$10^{-6}$
BER
0
31 dB
30 dB
22 dB
20 dB
static
1000
28
1
variation rate $S$
R1 direct
R2 frozen
R3 adaptive